Method to increase single-walled nanotube yield 15 times

Researchers at Skoltech University in Russia have achieved a significant breakthrough in the production of single-walled carbon nanotube (SWCNT) films. They introduced hydrogen gas alongside carbon monoxide during the chemical vapour deposition (CVD) process, resulting in a nearly fifteenfold increase in SWCNT yield without compromising quality. This addresses a major limitation in SWCNT manufacturing. Using various analytical techniques, the study examined how hydrogen impacted catalyst activation and nanotube growth. It identified two distinct temperature ranges where hydrogen had a beneficial effect: at 880°C, it increased yield by approximately 15 times, while at 1000°C, it extended SWCNTs by 2.4 times, reducing resistance and increasing yield.

Cost-effective, eco-friendly alternative to traditional solar cells

Researchers from Linköping University and the Royal Institute of Technology (KTH) have made a significant breakthrough in organic solar cells by incorporating untreated kraft lignin, a natural and abundant organic material, into the cells. This innovation addresses the environmental concerns associated with traditional silicon-based solar cells and the reliance on oil-derived polymers in organic solar cells. The researchers used kraft lignin extracted directly from wood pulp to create a component of the electron transport layer in the solar cell. This transition to sustainable wood-based materials in solar cell production can offer cost-effective and eco-friendly alternatives to traditional solar cells.

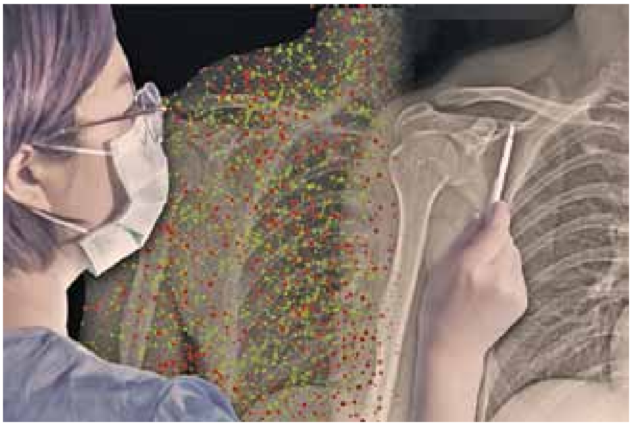
Organic solar cells (Credit: www.azom.com)

Twin-technology solar system with energy to power 753 homes 24x7

Researchers in Qatar and Jordan have proposed the twin-technology solar system (TTSS), an innovative system that combines solar updraft and downdraft technologies in a single unit, enabling it to generate electricity continuously both day and night. The system includes dual concentric inner and outer solar towers, turbines, water sprinklers, and a collector. The inner tower operates as a conventional solar updraft tower, while the outer tower uses a downdraft mechanism.

AI collaboration in medical imaging with personalised onboarding enhances accuracy and trust

Researchers at MIT and the MIT-IBM Watson AI Lab have developed a personalised onboarding system for radiologists using AI assistants to enhance their decision-making accuracy. This system provides guidance on when to trust AI guidance and improves collaboration between humans and AI. It resulted in a 5% accuracy improvement when humans and AI collaborated on image prediction tasks. The automated system learns from human-AI interactions and can be applied in various scenarios where humans and AI collaborate, including social media content moderation and content creation. The researchers believe that such onboarding will become essential for training medical professionals relying on AI in their field.



A new onboarding technique from MIT can help humans collaborate more effectively with AI assistants (Credit: Christine Daniloff, MIT; iStock)

Improved performance of thin-film VCSELs on composite metal substrate

Researchers at Yang Ming Chiao Tung University in Taiwan have successfully developed thin-film vertical-cavity surface-emitting lasers (VCSELs) operating at 940nm wavelength on a composite metal substrate. The substrate, made of copper/invar/copper (CIC), features a sandwich structure that allows for the transfer of VCSEL layers without cracking. Compared to traditional VCSELs on GaAs, the thin-film VCSELs on CIC substrate demonstrated lower voltage (1.37V vs 1.39V) and higher optical output power (24.40mW vs 21.91mW) at 1mA current. The CIC substrate's superior heat dissipation capabilities have the potential to significantly enhance the performance and thermal efficiency of thin-film VCSELs.